Alyson L. Singleton

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EDUCATION

Stanford University 2026 – present
Planetary Health Postdoctoral Fellow
Affiliations: Center for Human and Planetary Health, Center for Innovation in Global Health, London School of Hygiene & Tropical Medicine
Advisors: Giulio De Leo (Oceans and Earth System Science) & Marshall Burke (Global Environmental Policy)

Stanford University 2021 – 2026
Ph.D. in Environment and Resources (E-IPER)
Jim and Gaye Pigott Interdisciplinary Graduate Fellow
Thesis: *Global change, local risk: how development shapes infectious disease dynamics*
Fields of Inquiry: Global change ecology and infectious disease epidemiology
Advisors: Erin Mordecai (Biology) & Stephen Luby (Med - Infectious Diseases)

Centers for Disease Control and Prevention 2020 – 2021
ORISE Data Science Fellow
Advisor: Anna Blackstock (CDC/DDID/NCEZID/DFWED)

Brown University 2019 – 2020
M.A. in Biostatistics, Health Data Science Track
Advisor: Brandon Marshall (Epidemiology)

Brown University 2015 – 2019
B.S. in Applied Mathematics, Honors
Honors Thesis: *Network structure and rapid HIV transmission among people who inject drugs in Indiana: A simulation-based analysis*
Advisors: Matthew Harrison (Applied Mathematics) & Brandon Marshall (Epidemiology)

PUBLICATIONS

1. **Singleton, A. L.**, Lescano, A. G., Martel, K. S., MacDonald, A. J., Mandle, L., Sipin, T. J., Bautista, A. S. M., Carrera, E. D. R., Choque, G. A., Munayco, C. V., Luby, S. P., Mordecai, E. A. "Increased dengue transmission following highway paving in the Peruvian Amazon." *Nature Sustainability* (2026). https://doi.org/10.1038/s41893-026-01879-8
2. Anjos, R. S. D., Neves, N. D. S., Tuan, R., Palasio, R. G. S., Anjos, L. S. D., Kampel, M., Silva, V. A. F. D., Leo, G. A. D., **Singleton, A. L.**, Glidden, C. K., Chamberlin, A., Aslan, I. H., Pinter, A., Escada, M. I. S., Monteiro, A. M. V. "Assessing schistosomiasis transmission risk based on integrated hydrological and environmental modeling." *Environmental Research: Health* 4, no. 2 (2026): 025005. https://doi.org/10.1088/2752-5309/ae583d

3. Ferreira da Silva, V. A., Kampel, M., Silva dos Anjos, R., Gardini Sanches Palasio, R., Escada, M. I. S., Tuan, R., **Singleton, A.**, Glidden, C. K., Chamberlin, A., De Leo, G. A., Pinter, A., Monteiro, A. M. V. "Mapping schistosomiasis risk landscapes and implications for disease control: A case study for low endemic areas in the Middle Paranapanema river basin, São Paulo, Brazil." *PLOS Neglected Tropical Diseases* 18, no. 11 (2024): e0012582. https://doi.org/10.1371/journal.pntd.0012582

4. **Singleton, A. L.**, Glidden, C. K., Chamberlin, A. J., Tuan, R., Palasio, R. G., Pinter, A., Caldeira, R. L., Mendonça, C. L., Carvalho, O. S., Monteiro, A. M. V., Athni, T. S., Sokolow, S. H., Mordecai, E. A., De Leo, G. A. "Species distribution modeling for disease ecology: A multi-scale case study for schistosomiasis host snails in Brazil." *PLOS Global Public Health* 4, no. 8 (2024): e0002224. https://doi.org/10.1371/journal.pgph.0002224

5. Glidden, C. K., **Singleton, A. L.**, Chamberlin, A., Tuan, R., Palasio, R. G., Caldeira, R. L., Monteiro, A. M. V., Lwiza, K. M., Liu, P., Silva, V., Athni, T. S., Sokolow, S. H., Mordecai, E. A., De Leo, G. A. "Climate and urbanization drive changes in the habitat suitability of *Schistosoma mansoni* competent snails in Brazil." *Nature Communications* 15, no. 1 (2024): 4838. https://doi.org/10.1038/s41467-024-48335-9

6. Griffin, I., King, J.*, Lyons, B.*, **Singleton, A. L.***, Deng, X., Bruce, B. B., Griffin, P. M. "Estimates of SARS-CoV-2 Hospitalization and Fatality Rates in the Prevaccination Period, United States." *Emerging Infectious Diseases* 30, no. 6 (2024): 1144-1153. https://doi.org/10.3201/eid3006.231285

7. Turner, M. A., **Singleton, A. L.**, Harris, M. J., Harryman, I., Lopez, C. A., Arthur, R. F., Muraida, C., Jones, J. H. "Minority-group incubators and majority-group reservoirs support the diffusion of climate change adaptations." *Philosophical Transactions of the Royal Society B* 378, no. 1889 (2023). https://doi.org/10.1098/rstb.2022.0401

8. Benedict, K., **Singleton, A. L.**, Jackson, B. R., Molinari, N. A. M. "Survey of incidence, lifetime prevalence, and treatment of self-reported vulvovaginal candidiasis, United States, 2020." *BMC Women's Health* 22, no. 1 (2022): 1-9. https://doi.org/10.1186/s12905-022-01741-x

9. **Singleton, A. L.**, Marshall, B. D. L., Bessey, S., Harrison, M. T., Galvani, A. P., Yedinak, J. L., Jacka, B. P., Goodreau, S. M., Goedel, W. C. "Network structure and rapid HIV transmission among people who inject drugs: A simulation-based analysis." *Epidemics* (2020): 100426. https://doi.org/10.1016/j.epidem.2020.100426

10. **Singleton, A. L.**, Marshall, B. D. L., Zang, X., Nunn, A. S., Goedel, W. C. "Added benefits of pre-exposure prophylaxis use on HIV incidence with minimal changes in efficiency in the context of high treatment engagement among men who have sex with men." *AIDS Patient Care and STDs* 34, no. 12 (2020). https://doi.org/10.1089/apc.2020.0151

11. Marshall, B. D., Goedel, W. C., King, M. R., **Singleton, A.**, Durham, D. P., Chan, P. A., Townsend, J. P., Galvani, A. P. "Potential effectiveness of long-acting injectable pre-exposure prophylaxis for HIV prevention in men who have sex with men: a modelling study." *The Lancet HIV* 5, no. 9 (2018): e498–e505. https://doi.org/10.1016/S2352-3018(18)30097-3

**indicates equal authorship*

BOOK CHAPTERS

1. **Singleton, A. L.** "Infrastructure Development Shapes Infectious Disease Dynamics." In J. Drake & E. A. Mordecai (eds.), *Global Change and Emerging Infectious Diseases*. Oxford University Press. Forthcoming.

WORKS IN PROGRESS *(available upon request)*

1. **Singleton, A. L.**, Lana, R. M., Xavier, D. R., Vélez, J., Monteiro, A. M. V., Burke, M., Luby, S. P., Mordecai, E. A. Highways and dengue transmission in the Brazilian Amazon: linking infrastructure to disease occurrence and burden.
2. **Singleton, A. L.***, Vélez, J.*, Lana, R. M., Xavier, D. R., Matos, V. P., Luby, S. P., Burke, M., Mordecai, E. A. Geocoded point locations of CNES health facilities in Brazil.

CONFERENCES & OTHER PRESENTATIONS

** indicates poster presentation; otherwise talk was given*

2026	1st Latin American Congress on Road Ecology; AGU Annual Meeting*; Natural Capital Symposium, Stanford University; 15th Annual CEND Symposium: From Discovery to Delivery, UC Berkeley*; Center for Malaria and Vector-Borne Diseases, UCSF; 12th Stanford Global Health Research Convening; Stanford Doerr School of Sustainability Research Review
2025	Epidemics 10: International Conference on Infectious Disease Dynamics*; Celebrating Human and Planetary Health, Stanford University; Centro Nacional de Epidemiología, Prevención y Control de Enfermedades (CDC Perú); Ecology and Evolution of Infectious Disease (EEID)*; Stanford 3 Minute Thesis (finalist); 11th Stanford Global Health Research Convening*
2024	GISDay, Stanford University; Ecology and Evolution of Infectious Disease (EEID)*; Stanford Doerr School of Sustainability Research Review
2023	Stanford Data Science Conference*; Foley Lab, University of California, Davis
2022	XVI International Symposium on Schistosomiasis*; 8th Stanford Global Health Research Convening
2020	IDWeek*
2019	Epidemics 7: International Conference on Infectious Disease Dynamics*

MEDIA COVERAGE OF RESEARCH

"Highway Expansion Linked to Dengue Surge in the Amazon." *Think Global Health*.	Sep 2026
"El dengue viaja en carretera." *El País*.	Aug 2026
"Dengue surge linked to road development in Peru." *Stanford Report*.	Jul 2026
"Forecasting climate's impact on a debilitating disease." *Stanford Report*.	Aug 2024

GRANTS, FELLOWSHIPS, HONORS & AWARDS

Jim and Gaye Pigott Interdisciplinary Graduate Fellow (3-year tuition stipend) 2023 – 2026

The Stanford Interdisciplinary Graduate Fellowship (SIGF) Program is a competitive, university-wide program that awards three-year fellowships to outstanding doctoral students engaged in interdisciplinary research.

Stanford "3 Minute Thesis" Competition Finalist 2025

One of ten finalists selected across the University (YouTube link).

E-IPER Summer Research Grant ($4,000) 2022, 2023, 2025

Graduate McGee and Levorsen Research Grant ($2,000) 2022, 2025

Best Talk, Stanford Doerr School of Sustainability Research Review 2023

Award from the King Center on Global Development, Stanford Data Science Conference 2023

Spotlight Award for Creative and Effective Data Visualization, Stanford Data Science Conference 2023

Delta Omega Honorary Society in Public Health, Epsilon Iota Chapter, Brown University School of Public Health 2020

Honors in Applied Mathematics, Brown University 2019

FIELD EXPERIENCE

Pará, Brazil (three weeks) 2025

Visited highway corridors and interviewed local stakeholders across Pará with collaborators at Brazil's National Space Agency (INPE) and Oswaldo Cruz Foundation (Fiocruz). Supported vector trapping, drone surveying, and community engagement activities.

Lima, Peru & Madre de Dios, Peru (two weeks) 2025

Presented dengue and Interoceanic Highway findings to Peru CDC in Lima and to health and infrastructure authorities in Madre de Dios.

São Paulo, Brazil (two weeks) 2022

Visited *Biomphalaria* snail collection sites across São Paulo state with the Coordination for Disease Control of the State Health Secretariat (CCD-SP) and Fiocruz. Visited INPE's campus in São José dos Campos.

Madre de Dios, Peru (two months) 2022

Conducted epidemiologic and ecologic field research with the Regional Health Directorate of Madre de Dios (DIRESA) and the Emerge team at Universidad Peruana Cayetano Heredia (UPCH), including household surveys, blood, stool, and urine sampling, and host and vector trapping.

TEACHING & WORKSHOPS

Peru CDC, Lima, Peru. Led workshop on natural experiments, study design, and causal inference methods. https://github.com/alyson-singleton/DiD_tutorial — 2025

Stanford University, Ecology and Evolution of Infectious Disease in a Changing World (BIO2N), Guest Lecture — 2025

Stanford University, Human and Planetary Health (SUSTAIN 103), Guest Lecture — 2024

Stanford University, Ecology and Evolution of Infectious Disease Conference (EEID). Co-led workshop on Google Earth Engine for disease ecology applications. — 2024

Stanford University. Co-led workshop on species distribution modeling techniques for international collaborators and students visiting from Universidad Peruana Cayetano Heredia (UPCH). https://github.com/ckglidden/UPCH-species-distribution-tutorial — 2022

Stanford University, Global Change and Emerging Infectious Disease (EARTHSYS 214), Teaching Assistant — 2022

MENTORSHIP

Harper Baer (King Center Undergraduate Fellow and Biology Research Credits) — 2025 – 2026
Forest Fragmentation, Biodiversity Loss, and Zoonotic Disease Risk in the Amazon.

Raina Talwar Bhatia (Biology Summer Undergraduate Research Program, King Center Undergraduate Fellow, and International Policy Honors Thesis) — 2022 – 2026
Pave, Rain, Leptospira Gain: The Impact of Climate and Land-Use Change on Leptospirosis in Brazil 2007–2023. Awarded a 2025 Firestone Medal for Excellence in Undergraduate Research.

Kunal Arora (Science Writing Advancing Planetary Health Program) — 2024 – 2026
Co-creation of scientific communication products for Dengue and the Interoceanic Highway research project (*Think Global Health*).

Lyn Lee Loth (STEM Fellow and Biology Honors Thesis) — 2023 – 2026
Urbanization and Dengue Transmission in a Changing Cambodia.

Sage McGinley-Smith (Computer Science Departmental Independent Project and Computer Science Research Credits) — 2023 – 2024
Classification of pineapple and palm plantations in Costa Rica for environmental and human health applications.

REFEREE SERVICE

Nature Sustainability, Environmental Research, One Earth, Ecology Letters, Emerging Infectious Diseases, PLOS One, PLOS Global Public Health, PLOS Neglected Tropical Diseases, BMJ Global Health, AIDS and Behavior, AIDS Patient Care & STDs

LEADERSHIP & SERVICE

Session Co-Organizer, *The Hidden Health Costs of Infrastructure: Win-Win Nature-Based Approaches to Reduce Infectious Disease Risk*, Stanford Natural Capital Symposium	2026
E-IPER Executive Committee, Doctoral Student Representative, Stanford University	2024 – 2025
Scientific Planning Committee, Ecology and Evolution of Infectious Disease Conference, Stanford University	2024
Stanford Undergraduate Research Association Mentorship Program	2021 – 2023
E-IPER Social Committee, Student Leadership Council, Stanford University	2021 – 2022
Surveillance, Information Management, and Statistics Office representative and voting member, CDC DFWED Tactical and Mission Coordination Teams	2020 – 2021
Volunteer co-teacher, community college business math class, Rhode Island Department of Corrections	2019 – 2020

LANGUAGES

English (first language)

Spanish (full professional proficiency)